

GARDENCITY UNIVERSITY

ADVANCE JAVA

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SEM – 5TH, SEC – A

ROLL NO- 24BTCE126

PROGRAMS-

1. Write a Java program to demonstrate the concept of classes and objects. Create a class named Student with data members such as student name and student age. Create an object of the class and display the student details using a method.

Ans -



The screenshot shows the OneCompiler IDE interface. The main editor displays a Java file named Student.java with the following code:

```
1 class Student {  
2     String name;  
3     int age;  
4  
5     void displayDetails() {  
6         System.out.println("Student Name: " + name);  
7         System.out.println("Student Age: " + age);  
8     }  
9  
10    public static void main(String[] args) {  
11        Student s1 = new Student();  
12  
13        s1.name = "Sradha";  
14        s1.age = 20;  
15  
16        s1.displayDetails();  
17    }  
18 }
```

On the right side, the Console output shows the results of running the program:

```
Student Name: Sradha  
Student Age: 20
```

2. Write a Java program to demonstrate the concepts of inheritance and polymorphism. Create a parent class named Animal and a child class named Dog. Use method overriding to achieve runtime polymorphism and display the behavior of both classes.

Ans –

Animal.java × +

```
1 class Animal {
2     void sound() {
3         System.out.println("Animal makes a sound");
4     }
5 }
6
7 class Dog extends Animal {
8     @Override
9     void sound() {
10        System.out.println("Dog barks");
11    }
12 }
13
14 public class Main {
15     public static void main(String[] args) {
16         Animal a1 = new Animal();
17         a1.sound();
18
19         Animal a2 = new Dog();
20         a2.sound();
21     }
22 }
```

Console I/O AI Agent

Animal makes a sound
Dog barks